

AI IMAGE GENERATION USING PROMPT ENGINEERING

Objective:

To explore different AI-powered text-to-image generation tools and understand how prompts can be used to create AI-generated images.

AI Tools Explored:

Today, I explored and learned about various AI tools that can generate images from text prompts. I experimented with different tools to understand how the same prompt can produce different visual results.

Text-to-Image AI Tools:

- ❖ Meshy AI
- ❖ Adobe Firefly
- ❖ Leonardo AI
- ❖ Canva AI
- ❖ Ideogram

What I Learned:

Through this exploration, I learned how different AI image-generation tools interpret text prompts and transform them into visual content. I also understood that the quality and structure of a prompt can influence the generated image.



MESHY AI

Prompt Used:

A spectacular futuristic Generative AI and Large Language Model (LLM) learning laboratory, designed as a high-end 3D educational technology environment for a B.Sc. Computer Science with Artificial Intelligence student. At the centre, create a powerful glowing artificial intelligence core shaped like a futuristic digital brain, surrounded by interconnected neural network circuits, transformer architecture components, flowing data streams, and holographic AI interfaces. Include a modern workstation with a powerful computer, laptop, keyboard, and multiple holographic displays showing Python programming code, prompt engineering workflows, large language model concepts, chatbot conversations, natural language processing, machine learning algorithms, AI-generated content, embeddings, and tokens.

Generator Image:



Adobe Firefly

Prompt Used:

Create a premium cinematic 2D digital artwork of a confident young female CEO named Swathi leading and mentoring a group of college students in a modern corporate interview and leadership presentation setting. Swathi is seated at the centre of a professional executive desk, confidently interacting with the students as if she is conducting an inspiring interview and discussing an innovative company project. She is wearing an elegant professional black outfit with long, naturally flowing loose hair. Her expression is confident, intelligent, friendly, and inspiring. Place a professional studio microphone on the desk directly in front of her. Add a clean professional nameplate on the desk that clearly reads "SWATHI" and "CEO-SIVA FOUNDATION" Behind Swathi, place a large modern digital presentation screen displaying professional business analytics, bar charts, line graphs, project growth statistics, and futuristic data visualisations related to artificial intelligence, technology, education, and business innovation. The students should be seated around the table facing Swathi, listening carefully and participating in the discussion. Show a collaborative, inspiring atmosphere where the CEO guides the students on their project, technology, innovation, and future career opportunities. Set the scene inside a high-end modern technology foundation headquarters with glass walls, elegant furniture, subtle LED lighting, and a clean professional interior. Use cinematic lighting, realistic facial expressions, detailed clothing textures, natural body proportions, sophisticated composition, depth of field, soft reflections, and premium corporate aesthetics. Camera angle: slightly wide cinematic shot, with Swathi as the main focal point while keeping the students, microphone, desk, and presentation screen clearly visible. Overall mood: professional, innovative, inspirational, powerful, modern, futuristic, and welcoming. Image style: high-quality 2D cinematic digital art, polished professional corporate campaign aesthetic, high detail, sharp focus, balanced composition, landscape 16:9. Avoid distorted faces, extra fingers, unnatural body proportions, blurry subjects, excessive text, watermarks, random logos, and cluttered backgrounds.

Generator Image:



Leonardo AI

Prompt Used:

Create a highly realistic cinematic lifestyle photograph of a beautiful young woman sitting beside a large glass window inside a stylish modern café on a peaceful rainy evening. She is the main focal point, with naturally flowing slightly wavy dark hair, realistic facial features, expressive eyes, natural skin texture, and a calm, confident, thoughtful expression. She should look naturally attractive and elegant, without excessive makeup or artificial beauty filters. She wears a sophisticated casual outfit with a neutral top, stylish long coat, and minimal accessories. She sits at a wooden café table with an open laptop, coffee cup, notebook, pen, and books, working on a creative digital project while looking thoughtfully toward the rainy window. Outside, show a realistic modern city street with raindrops on the glass, wet pavement, glowing street lights, blurred city lights, distant pedestrians with umbrellas, and subtle vehicle headlights. Inside, include warm pendant lights, wooden furniture, indoor plants, minimalist décor, and a few distant customers. Create a beautiful contrast between warm café lighting and cool rainy tones outside. Use professional lifestyle photography, natural perspective, realistic depth of field, soft bokeh, cinematic lighting, realistic shadows and reflections, detailed hair and clothing textures, and balanced exposure. Position the woman slightly to one side with natural leading lines from the window and table. Style: photorealistic, sophisticated, authentic, premium editorial photography, cinematic, high detail, natural beauty, professional composition. Mood: peaceful, intelligent, confident, creative, inspiring, warm, and modern. Avoid cartoon or anime styles, plastic skin, excessive makeup, distorted faces, extra fingers, deformed hands, unnatural proportions, blurry faces, unrealistic hair, excessive saturation, text, logos, watermarks, and cluttered backgrounds.

Generator Image:



Canva AI

Prompt Used:

Create a highly realistic cinematic night photograph of a beautiful young woman playing gently with a mother dog and three adorable puppies on a quiet city street at midnight. The scene should feel warm, emotional, peaceful, natural, and authentic. The woman has long loose dark hair and wears a stylish casual outfit suitable for a cool night. She is sitting or crouching comfortably near the dogs, smiling naturally while interacting playfully with the puppies. The mother dog should stay close to her puppies, creating a warm family atmosphere. The street is mostly dark, illuminated naturally by a few beautiful warm streetlights. Add subtle reflections on the pavement, soft shadows, distant city lights, and a slightly blurred nighttime background. Keep enough light on the woman and dogs so their faces and details are clearly visible. Show three different puppies playing naturally around the woman while the mother dog watches them. Make the dogs look anatomically realistic with natural fur, expressions, body proportions, and playful movements. Use cinematic photography, realistic skin texture, detailed hair, realistic animal fur, natural lighting, shallow depth of field, soft bokeh, realistic shadows, and professional composition. The woman and dogs should be the main focus. The atmosphere should communicate kindness, friendship, comfort, and the beautiful connection between humans and animals. Style: photorealistic, cinematic night photography, natural, emotional, premium lifestyle photography, high detail, realistic lighting. Aspect ratio: 4:5 portrait. Avoid cartoon style, anime, artificial faces, distorted hands, extra fingers, unrealistic dogs, duplicate puppies, blurry subjects, excessive darkness, oversaturated colors, text, logos, and watermarks.

Generator Image:



RESULTS AND CONCLUSION

Results

I explored five AI image-generation tools: **Meshy AI, Adobe Firefly, Leonardo AI, Canva AI, and Ideogram**. I learned that clear and detailed prompts help AI tools generate images closer to the intended concept. I also observed that different tools can interpret the same prompt differently.

Key Learning

- Learned the importance of structured prompts.
- Understood how prompts affect image quality.
- Explored different AI image-generation tools.
- Gained practical experience in prompt engineering.

Conclusion

This task helped me understand the practical application of **Prompt Engineering in Generative AI**. By exploring different AI image-generation tools, I learned how written instructions can be transformed into creative and realistic visual content.

I understood that an effective prompt should clearly describe the required subject, setting, visual style, lighting, composition, mood, and other important details. I also learned that improving and refining a prompt can help achieve a more suitable output.

Overall, this activity gave me valuable practical experience with **AI-powered image generation and prompt engineering**. It improved my understanding of how different AI tools respond to prompts and how structured prompts can be used to create more relevant and visually appealing images.

Overall Outcome

AI Tools Explored: 5

Main Skill: Prompt Engineering

Application: Text-to-Image Generation

Learning Outcome: Understanding how structured prompts influence AI-generated images.